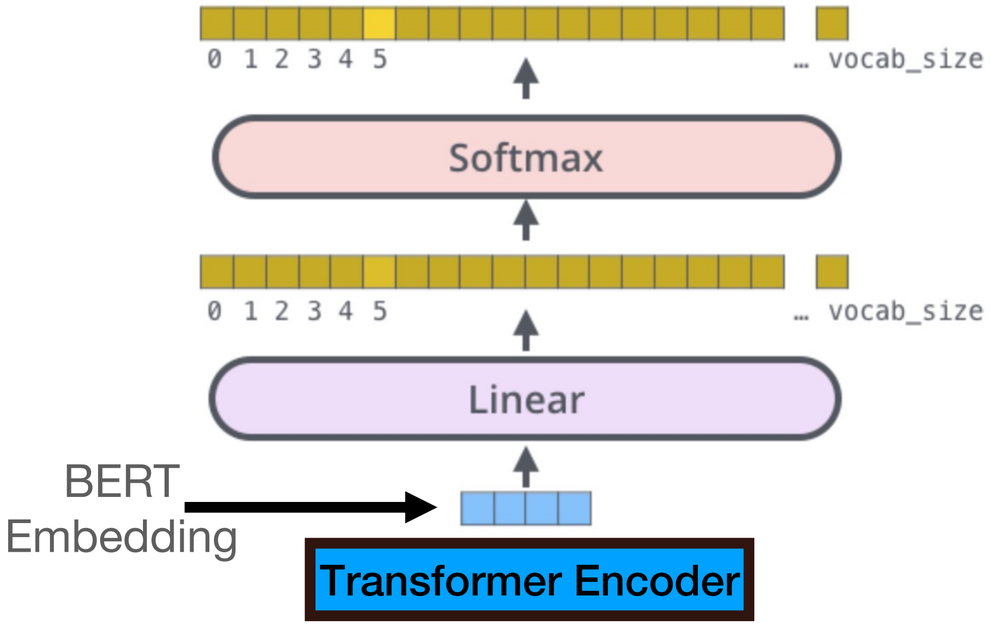
The BERT Model

- The first widely popular model to use pre-training
- The BERT model is a standard model for obtaining contextualized word embeddings
- The model is built like a *Transformer* **encoder**, except that instead of generating a list of vector embeddings, it feeds these vectors to a linear layer, followed by a soft-max
- BERT provides embeddings for each token in the input sequence – the computed vector that enters the linear layer



BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding
https://arxiv.org/abs/1810.04805

Masked Language Models

- BERT is trained as a **masked language model** (MLM)
- MLMs are statistical models that, given a sentence where part of the tokens are replaced with masks, predict the identity of the masked tokens

My dog is [MASK] and likes to [MASK] the entire [MASK].

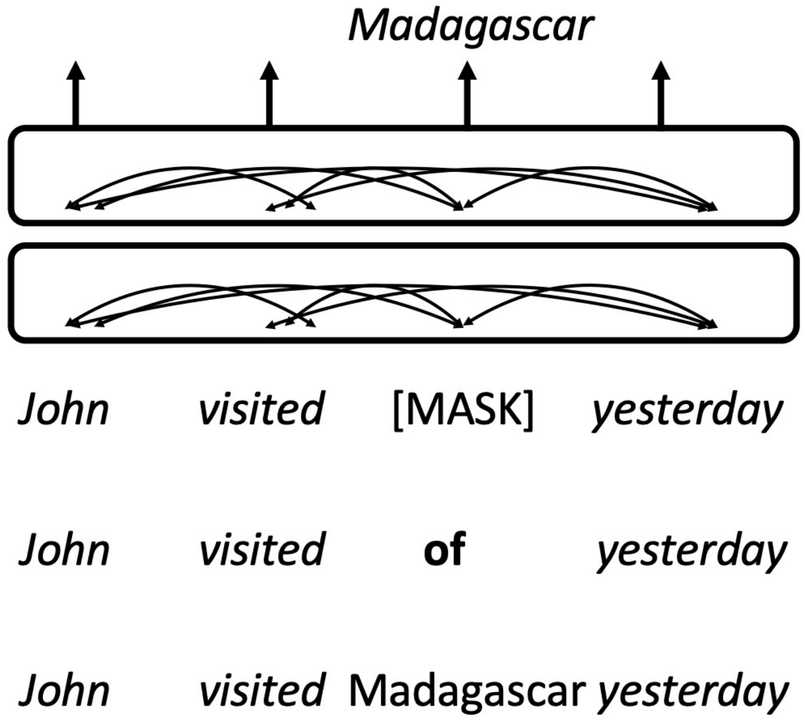
Mask 1

Prediction	Score
My dog is **hungry** and likes to *[MASK2]* the entire *[MASK3]* .	5.6%
My dog is **friendly** and likes to *[MASK2]* the entire *[MASK3]* .	4.8%
My dog is **cute** and likes to *[MASK2]* the entire *[MASK3]* .	3.7%
My dog is **nice** and likes to *[MASK2]* the entire *[MASK3]* .	2.9%
My dog is **smart** and likes to *[MASK2]* the entire *[MASK3]* .	2.4%

https://demo.allennlp.org/masked-lm

BERT's Training

- BERT's training is carried out by segmenting texts into windows in the size of the model's input window, and selecting 15% of the tokens (denote the set with *S*)
- The tokens in *S* are altered such that:
  - 80% of them are replaced with the special token <MASK>
  - 10% of them are replaced with a random token
  - 10% of them remain the same
- Denote the modified input sequence with ***x'***

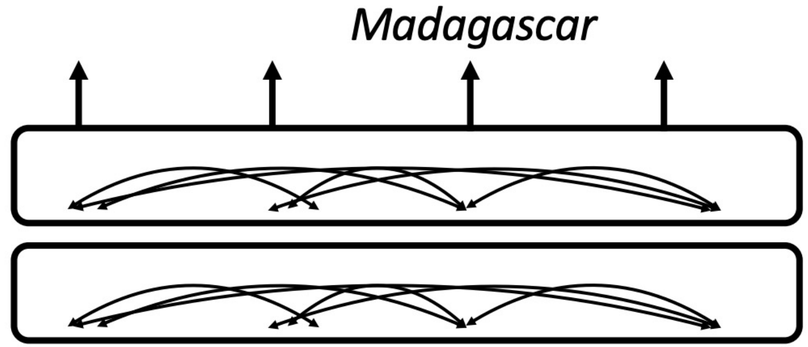


BERT's Training

- The loss function is then the cross entropy loss over the tokens in S:

$$L(\theta) = -\frac{1}{|S|} \sum_{i \in S} log(p_\theta(x_i|x'))$$

- During inference, no masking is employed



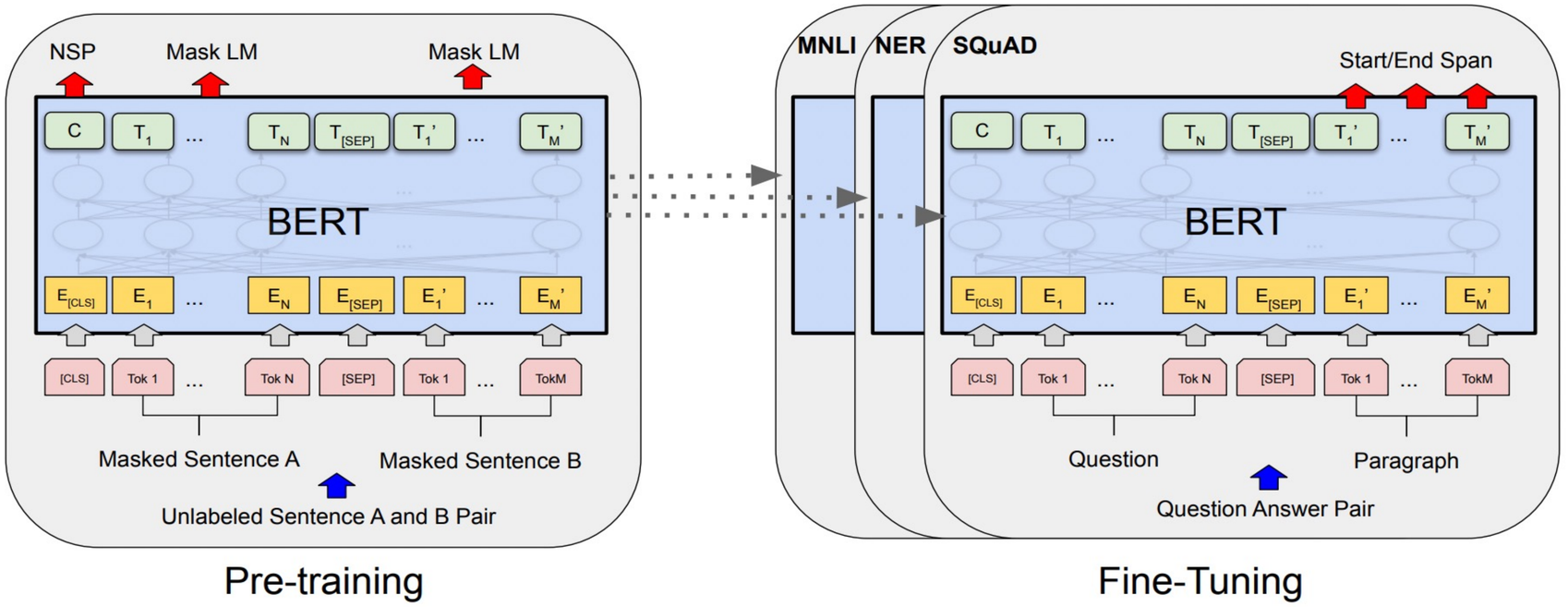
John *visited* [MASK] *yesterday*

John *visited* **of** *yesterday*

John *visited* Madagascar *yesterday*

Fine Tuning BERT

- The standard paradigm for using BERT in NLP includes two phases:
  - Pre-training
  - Fine-tuning



Examples for fine-tuning BERT

- POS tagging:
  - Often viewed as an independent prediction task (not structured prediction)
  - A classification head for each word
    - Its input is the BERT embedding and its output is a soft-max in the dimension of the number of POS tags.
  - As usual, the *j*-th coordinate of the soft-max is interpreted as $P(y_j|x_1,\ldots,x_n)$
  - Training set:
    - Each sample consists of the sentence $(x_1,\ldots,x_n)$, index *i*, and gold POS tag for *i*-th token y_i^*
    - Cross entropy loss over $P(y_i^*|x_{1,\ldots,n})$

Examples for fine-tuning BERT

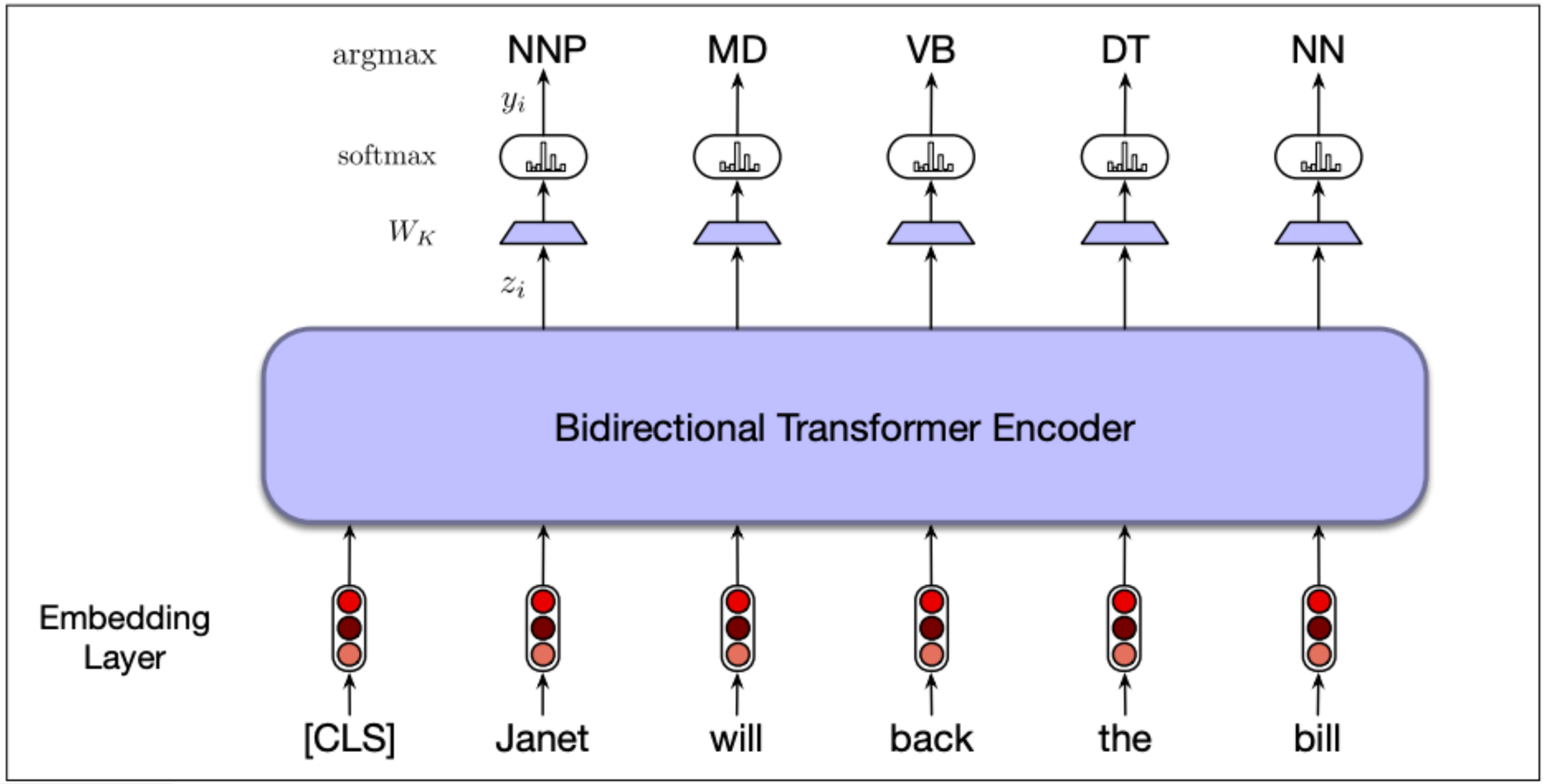


Figure 11.9 Sequence labeling for part-of-speech tagging with a bidirectional transformer encoder. The output vector for each input token is passed to a simple k-way classifier.

Next Sentence Prediction

- BERT is in fact pretrained not only as a masked language, but also as a model for Next Sentence Prediction (NSP)
- NSP is the task that requires, given two sentences, to determine whether the two sentences form a sensible consecutive pair or is just two random sentences
    - NSP training data: positive - pairs of adjacent sentences, negative - pairs of random sentences

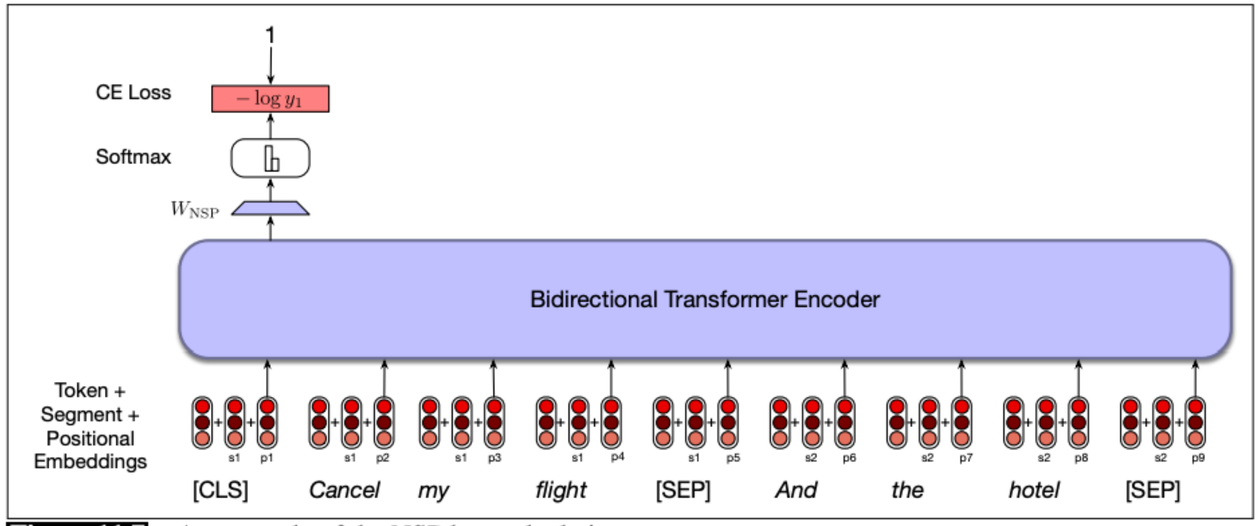


Figure 11.7 An example of the NSP loss calculation.

Next Sentence Prediction

- Each training sample is prepended with a special [CLS] token. A special [SEP] token is added between the sentences
- The loss function is over the [CLS] where the labels are 1/0
- [CLS] is often used for fine-tuning on sentence classification tasks

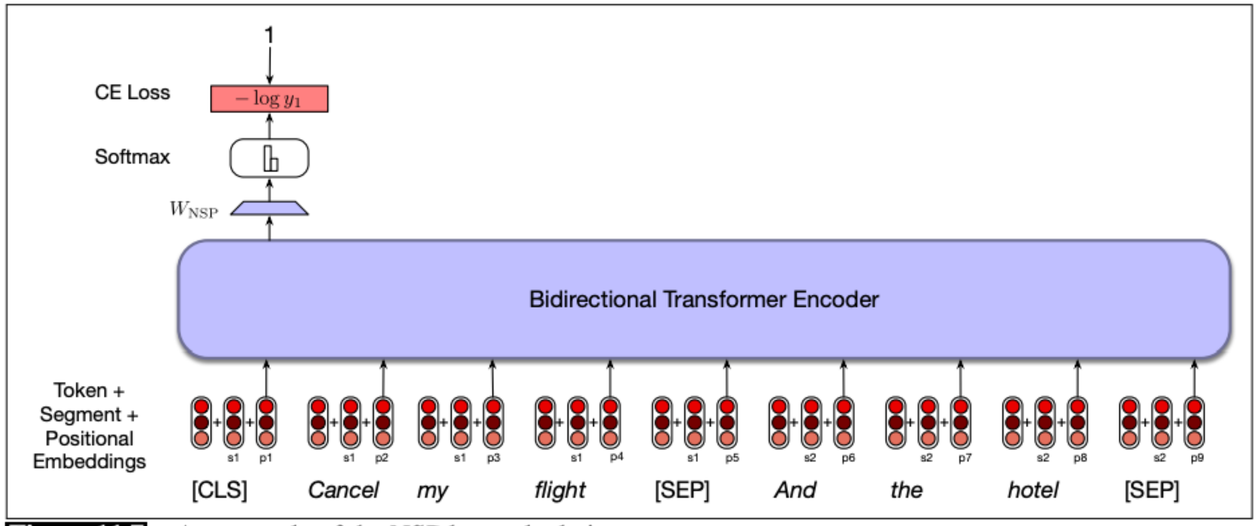


Figure 11.7 An example of the NSP loss calculation.

Examples for fine-tuning BERT

- Sentiment analysis:
  - Viewed as a sentence classification task
  - A classification head is placed over a special token ([CLS])
    - As before, usually a linear layer followed by a soft-max
  - The soft-max is interpreted as $P(class|x_1,\ldots,x_n)$
    - *class* takes values in the set of potential sentiment classes
  - Training set:
    - Each sample consists of the sentence $(x_1,\ldots,x_n)$, gold POS tag for the sentence y^*
    - The loss function is the expectation over $log\ P(y^*|x_1,\ldots,x_n)$

Results of Fine-tuning BERT

- BERT achieved very impressive results across a range of token and sentence classification tasks, sequence prediction, as well as other baselines
- A very broad experimental setup in terms of task types and baselines
- It was widely adopted, and is still often used (about 5 years after its inception)
- For detailed results, see the original BERT paper